Fahad Hamza Minhas

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PROFESSIONAL SUMMARY

Final-year Computer Science student at University of Management and Technology, specializing in full-stack MERN development and deep learning-based computer vision systems. Experienced in building scalable web applications and integrating YOLO-based detection and segmentation models into production-ready healthcare solutions.

EDUCATION

University of Management and Technology (UMT) — Lahore, Pakistan
Bachelor of Science in Computer Science — Sept. 2022 – July 2026

- **Relevant Coursework**: Deep Learning, Advanced Web Technologies, Data Structures & Algorithms. Object Oriented Programming

PROJECTS

MedVision AI (Final Year Project) — Dec. 2025 – Present

- Developing a full-stack brain tumor detection system using the **MERN Stack** and **YOLO11-seg** model.
- Engineered a secure MRI upload pipeline using Node.js; optimizing the data flow between the web interface and the YOLO model
- Achieved a **96.6%** Intersection-over-Union (IoU) score on validation data for tumor segmentation.
- Building a responsive dashboard for clinicians to visualize patient results and manage diagnostic history.

Tech Stack: React.js, Next.js, Node.js, Express.js, MongoDB, Python, YOLO-based segmentation model.

MERN Stack CRM Dashboard — 2025

- Architected and developed a real-time CRM dashboard using **React.js** and **Node.js** to manage users and customer records.
- Designed RESTful APIs using Express.js and MongoDB for efficient CRUD operations and persistent data handling.
- Implemented global state management using React Context API to dynamically update UI components without page reloads.

Tech Stack: React.js, Node.js, Express.js, MongoDB

TECHNICAL SKILLS

Programming Languages: JavaScript (ES6+), Basic TypeScript, Basic Python, C++, HTML5, CSS3
Web Development: MERN Stack, Next.js, React.js (Hooks, Context API, Zustand), RESTful APIs.
AI/Machine Learning: Deep Learning, Computer Vision, YOLO (v11-seg), Recurrent Neural Networks (RNNs)
Databases: MongoDB (NoSQL), MySQL, Database Design Management
DevOps& Tools: Git/GitHub, Postman, VS Code

ACHIEVEMENTS

Maintained a **3.71 CGPA** throughout undergraduate studies at University of Management and Technology.

Selected for Final Year Project focused on AI-driven medical image segmentation (MedVision AI). Completed multiple academic projects integrating full-stack development with machine learning systems.